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Relative price and deterrence: the effect of speeding tickets on road behavior

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Abstract: Governments use fines in order to deter socially unwanted behavior on a wide array of issues. It is commonly assumed that, fixing the probability of sanction, the higher the price, the more deterrent a sanction is. I use data from automated speed cameras in Bogotá, Colombia to measure the deterrent effect of a speeding ticket. By using the real speed limit as a source of exogenous variation and a regression discontinuity design, I find that speeding tickets have a general deterrent effect on speeding behavior but a null effect on reducing traffic accidents. Using the commercial price of the vehicle as a proxy for wealth and the fact that the monetary sanction is fixed, I find that the deterrent effect is concentrated on the higher quintiles of the distribution. I validate these findings with a judge-IV using the fact that a police agent manually verifies all tickets. Other dimensions of driving behavior show that speeding tickets do change behavior, not only adjust beliefs about enforcement. In short, speeding tickets might be less useful for preventing road fatalities than it is commonly assumed and inefficiently expensive for changing behavior for a considerable number of drivers. They also seem to be of little effectiveness for the current public health challenge of traffic fatalities in low and middle income countries. This evidence points to income based fines as a necessary public policy tool.

1. Introduction:

A central question for governments is how to improve the deterrent power of their legal sanctions. A common avenue is to increase the monetary sanctions under the assumption that a higher expected cost results in a lesser prevalence of unwanted behavior. In contexts with little State capacity for enforcement this assumption might be wrong; legal sanctions can be too expensive for certain actors that not only avoid paying the fine but also continue with the unwanted behavior.

This article starts by answering questions on the general causal effect of speeding tickets on behavior: What is the causal effect of a speeding ticket on subsequent behavior of a driver? The most studied subsequent behavior usually speeding itself, but with the data available I can also observe the chances of being involved in an accident. Since speeding tickets are used as a tool for preventing road accidents, this is the most relevant variable to study¹. After this general effect has been established, the article explores the heterogeneous effect by income of the driver to answer whether relatively more expensive sanctions are better or worse for deterring speeding and accidents. Finally, I explore whether the results can be attributed to adjustment of beliefs about law enforcement or behavioral change.

This paper relates to the literature on the analysis of the effects of financial fines and fees imposed by the State. There is ample public debate on the capacity of higher fines to increase revenue and the effect this has on recidivism. Some papers find a modest effect on revenue at the cost of higher recidivism (Giles, 2023). Others argue that even if the effect of sanctions is higher revenue, they have no effect on recidivism and thus are not justifiable ways of increasing deterrence (Finlay et al., 2024). The authors argue that the null effect on all relevant criminal behaviors is precisely estimated, not to lack of power. In recent work, (Norris & Rose, 2024) argue that reducing legal financial obligations (LFOs) will impact municipal revenues without increasing deterrence. The evidence provided by (Bing et al., 2024) on a randomized payment of existing fines for criminals shows that this policy only reduces recidivism related to the financial costs of the LFO itself. These effects are not negligible and point to no deterrent effect at a significant cost of engagement with the criminal system. In summary, LFO's seem to have a null effect on criminal behavior, while a modest effect on increasing municipal revenue. Questions of fairness arise from these two facts. In fact, this paper tries to bring questions of fairness (Khaneman et al., 1986) to the economic understanding of crime, using speeding tickets only as a data-rich case study to ask if smaller fines that can be perceived as fairer might be more effective at deterring behaviors socially undesired behaviors. The question of whether a fine is just like any other price (Gneezy & Rustichini, 2000), and what type of price it is (Feldman & Teichman, 2008) has been explored in many dimensions, but not specifically in the deterrence goal of many legal sanctions.

The question of the optimal amount of a fine has also been examined in the specific case of traffic tickets, but mainly through the lens of revenue, not deterrence like I do here. Some have shown that increasing fines has only a modest effect on timely payment (Traxler & Dušek, 2023) and recidivism (Dušek & Traxler, 2022). A very detailed of the effect of the price of ticket fines is (Kaila, 2024), who uses the relative to income price of traffic tickets in Finland to shows that there is in fact a reduction

¹ I will argue that studying the effect of traffic tickets on subsequent speed tickets is like measuring the effect of tacking a pill to reduce cholesterol on taking the next pill, not on cholesterol levels.

on recidivism for higher fines, but the effect is short lived and the fines need to be significantly more expensive. In their work, (Goncalves & Mello, 2024) also study the effect of harsher sanctions by exploiting the discretionary sanctions given by police officers. They find that harsher fines do reduce recidivism and, to a lesser degree, crash involvement. They also find that officers discretion is inefficient since harsher sanctions are given to drivers that are less deterred by them. Contributing to this new evidence on the indirect deterrent effect of traffic fines on road safety is one of the central goals of this paper.

Another branch of the literatures studies what citizens learn by interacting with law enforcement. Using a change on the punishment for recidivism, (Philippe, 2024) finds that learning about criminal law appears to be limited to people with direct involvement with the law. In the same direction, it has been shown that individuals do adjust their priors after receiving a speeding ticket (Dušek & Traxler, 2022). As I will explain later, I contribute to this literature by studying subsequent behavior where there is going to be future interactions with the law and also where there is not. I can speak to how interactions with the law affects behavior when citizens know the law is not present.

Finally, this paper also contributes to a broader literature on the effect of traffic enforcement on behaviors (De Angelo & Hansen, 2014; Hansen, 2015; Luca, 2015; Makowsky & Stratmann, 2011). In particular, my works add to a growing literature that studies fines and enforcement in relation to factors that do not affect either sanctions or probability of enforcement but have been shown to have an effect on traffic accidents. Things like the political cycle (Bertoli & Grembi, 2021), or the stock market (Fry & Farrell, 2023) (Giulietti et al., 2020). Close to this strand is a set of papers that explores behavioral responses to traffic enforcement and tries to derive policy recommendations to reduce road accidents and fatalities (Zhang et al., 2020) (Lu et al., 2016). My work contributes by suggesting perceptions of fairness of sanctions as a mediator of changes in enforcement. I also contribute to this literature by adding evidence from a city in the developing world, something that is still rare in the field.

Examining the effect of speeding tickets on behavior is difficult because of selection. Drivers that decide to exceed speed limits might be significantly different in unobservable characteristics that also relate to risky behavior, thus comparison between ticketed and not ticketed drivers cannot capture the causal effect of tickets. But in the case of Bogotá, the administrative process for giving speed tickets offers a way to solve this issue. Speed tickets are not given to all vehicles that exceed the maximum speed posted on road signs, but to a different speed with an additional tolerance. Since the number of kilometers per hour of this tolerance is not known to drivers, thus a discontinuity exists. A regression discontinuity design will be used to make a valid comparison between only drivers that decide to exceed the speed limit, but some receive a ticket and some do not. Evidence for the validity of the design will be provided on the methodology chapter of this paper. For examining the

motivating hypothesis of this work, that legal monetary sanction can be inefficiently expensive, I use the market price of the vehicle as proxy for the driver's income. In this way I can turn a fixed price into a price that is relative to the driver's income.

The paper proceeds as follows. Section 2 presents the institutional background of speed fines in Bogotá, Colombia and the Data that will be used on the estimation. Section 3 discusses the estimation strategy and presents the evidence necessary for its validity. Section 4 presents all the results from the estimation on the general effect of speeding tickets with a fixed price. Section 5 presents results on the relative price of vehicles. Section 6 presents some robustness checks on the main estimation. Section 7 briefly discusses the behavioral mechanisms that could be driving the results. Finally, section 8 discusses the main findings in relation to fines as an effective deterrent strategy and derives possible policy recommendations for its improvement.

2. Institutional background and data:

Automated enforcement of speed started in Bogotá in 2020, with the staggered installation of Cameras in the most dangerous zones. The program was originally called *Cámaras Salvavidas* (Spanish for Lifesaving cameras) to emphasize that the objective of the program was not municipal revenue but behavioral change. After some debate on the legality of the fines, Colombia's higher court² allowed for the fine to be given to the owner of the vehicle. Two elements of the legal frame are crucial in the rest of the paper. First, tickets have a fixed price independent³ of the type of vehicle or the income of the owner, or the speed at which a vehicle is circulating. Second, there is no additional fine for recidivism.

The central database I will use for defining treatment includes all vehicles that exceeded the legal speed limit under a Speed Camera from August 16 to October 15, 2024. This dataset is comprised of 181,489 distinct vehicles that passed the speed Cameras at the time at which they give speeding tickets, which corresponds to daylight. Since I will only be using vehicles that passed under the Cameras once a total of 158,703 comprise the full sample used for estimation⁴. This is equivalent to 87% of the plates that exceeded the limit. I will call this sample my Fulls sample. On the Appendix section 2 shows some results using also some vehicles that passed more than once.

² This is ruling C-321 of 2022 by Corte Constitucional del Colombia.

³ Technically there is an early payment, but all vehicles are subject to this. So, big or small the price is the same for everyone.

⁴ Full distribution of the number of observation by vehicle in this sample is shown on Table 1 of the Appendix. This is a particular problem since vehicles might circulate one time below the real limit and another one above the real limit, thus making it harder to classify them as treated or untreated in terms of the discontinuity. I will do robustness checks on this, but mostly just use vehicles that circulated once.

I can also access the databases for all existing tickets and accidents occurring in Bogotá⁵. I can see if each plate received a ticket or participated in an accident both before and after the period of study, until March 15 of 2025. Those two will be my main outcome variables but also are important in correcting for specific characteristics of the sample.

I can access the registration record for all vehicles in Bogotá (and some that pay congestion pricing⁶) and match the plate with information on brand, model and year of the vehicle. A web scrapping procedure of the most used website for selling used cars in the city is done to obtain the commercial value of each brand-model-year combination⁷. This leaves me with a total of 67,235 vehicles for which I can find the commercial prices, naturally a subsample but is big enough for the estimation to have statistical power. I will use the price of the vehicle as a proxy for wealth. The heterogenous effect of the relative price of the ticket can thus be estimated. Local effects and external validity questions will be analyzed in the Results section. I will call this sample my Matched sample.

In no way am I arguing that my Matched sample is similar to the Full Sample. The decision to register the vehicle in Bogotá is surely related to behavior as the fact that the registry is growing in surrounding municipalities but not in Bogotá. This is particularly caused by the decision of new motorcycle owners to register their vehicle outside Bogotá, even if this is where they plan to use it. Thus, the effect seen on that sample might only extend to other drivers that decide to register their vehicles in Bogotá.

3. Estimation strategy.

My estimation derives from a fact that is unknown to drivers, that tickets are not given at the speed posted on the road. On the following I will call the legal limit posted on the road sign (50 km/h) as the posted speed limit, and the limit at which most tickets really given (55 km/h) as the real limit⁸.

This discontinuity allows for some exogeneous variation that can be exploited. In most of my applications I will estimate the following equation:

$$Result_i = f(z_i) + \tau D_i + \epsilon_i \quad (1)$$

⁵ For the accident data I also use the data for the centralized health system that is specifically dedicated to traffic accidents. This data is under a system called SIRAS, and belongs to the national Minister of Transportation, not the city of Bogotá.

⁶ Some vehicles can pay to avoid a circulation restriction that exists in the city (*Pico y placa Solidario*). All the data I need is registered on these transactions so I can access the data in this case.

⁷ All data on vehicle prices is taken from tucarro.com, the most used website for selling used vehicles.

⁸ In the sample there is only one Camera that has 30 as the maximum speed. It only accounts for 4,460 observations (1.82%). Speeds are adjusted accordingly.

Where $Result_i$ is one of my three behaviors of interest for vehicle i : a dummy that takes the value of 1 if the vehicle had at least one instance of excess speed⁹ in the six months after receiving a ticket and 0 otherwise, a dummy that takes the value of 1 if the vehicle had one accident in the six months after receiving a ticket and 0 otherwise, or the total number of instances of excess speed in the six months after receiving a ticket.

$f(z_i)$ is a polynomial of degree 2 (Gelman & Imbens, 2019) for the variable that decides treatment, the speed at which vehicle i is circulating under a Speed Camera. All vehicles in the sample exceed the posted speed limit, but only some exceed the real limit. This reduces selection and allows me to identify causality.

D_i is my treatment variable, which takes the value of 1 if the vehicle received a speeding ticket and 0 otherwise. The treatment is instrumented via the assignment, since this is a fuzzy design, and the discontinuity at the real speed limit is around an increase of 27 p.p. on the probability of receiving a ticket.

ϵ_i is an error term.

τ is my coefficient of interest. I will argue that it captures the causal effect of receiving a ticket in subsequent behavior. It is a local effect, since it only captures the effect of a ticket for vehicles around the cutoff, but those are the vast majority of observations in the sample¹⁰. For all estimations I use the `rdrobust` package in STATA with the option to ignore mass points, as the variable that assigns treatment is discrete. Three estimator, conventional, bias-corrected and robust will be shown.

3.1 Validation of the Fuzzy Regression Discontinuity Design: Existence of discontinuity:

Figure 1 shows that enforcement does work with the real limit, not the one posted on the street signs. It also shows that the discontinuity is not 0 to 1, so my strategy will be a fuzzy design. In short, even if the behavior of driving at 55 km/h is very similar to driving at 56 km/h, and both are above the legal limit, the latter has four times the probability of receiving a ticket than the former.

⁹ Since the real limit is unknown to drivers, a future excess speed is the one at above the posted limit; that is 50 km/h, not 55 km/h.

¹⁰ 80% of observations pass under Speed Cameras at speeds between 51 and 60 km/h.

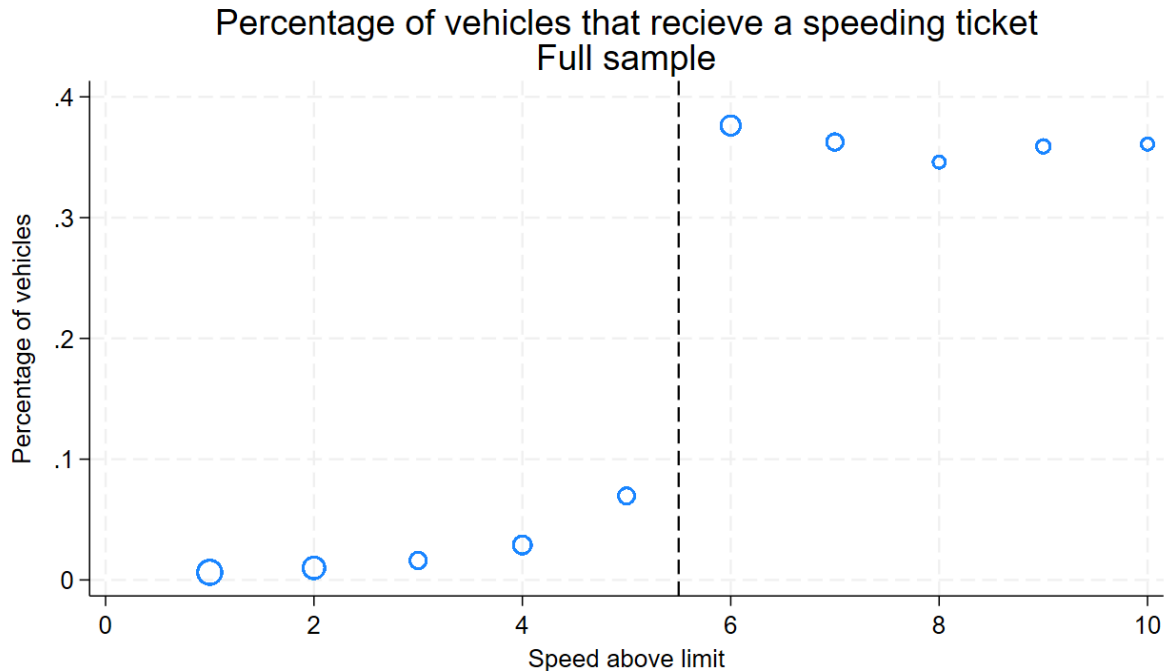


Figure 1. Speed above limit is constructed as: (vehicle observed speed-speed limit).

It is important to show that this discontinuity also exists on the matched sample and that is similar for different values of vehicle prices. Figure 2 plots this discontinuity for all matched vehicles and for each quintile of vehicle price. In fact, the discontinuity is slightly bigger in this case across the board, but particularly for higher quintiles.

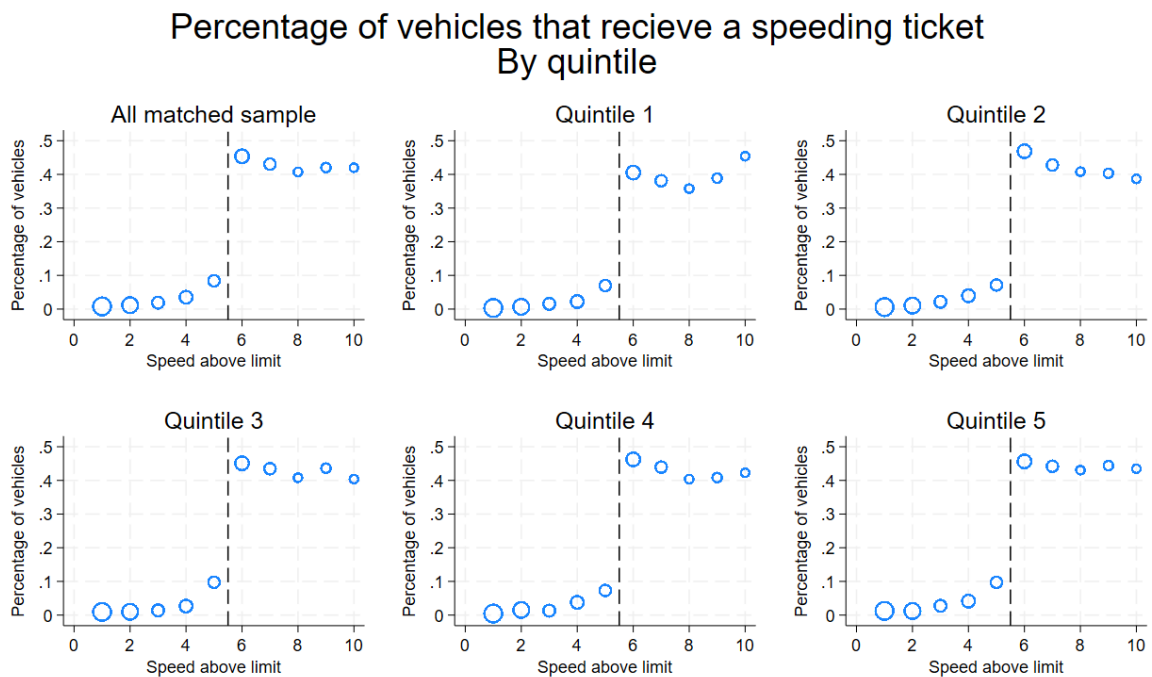


Figure 2. Speed above limit is constructed as: (vehicle observed speed-speed limit).

A very relevant question for local public policy is whether this deterrent effect of tickets varies by vehicle type. This is caused by motorcycles being a growing percentage of road injuries and fatalities both in Bogotá and in all Colombian cities, as in similar middle income countries. Figures 2 and 3 of the Appendix show this for the full and matched sample. It can be seen that the discontinuity is smaller for motorcycles, and that this is considerably smaller in the case of the matched sample. This is less a threat for the identification strategy and more of a fact to have in mind when interpreting results.

3.2 No manipulation.

Now that I have shown that there is a discontinuity at cutoff, I need to show that there is no manipulation. In short, drivers are not aware of this fact. Since the estimation wants to separate the effect of a possible ticket from any other change, I will only use the vehicles that have not received a ticket in the 90 days before passing under a Camera exceeding the legal speed limit. Frequency is shown in Figure 3. In the Appendix the same distribution is shown for all vehicles, independent of receiving or not receiving a ticket in the prior 90 days. It seems that drivers are not selecting themselves into not receiving a ticket.

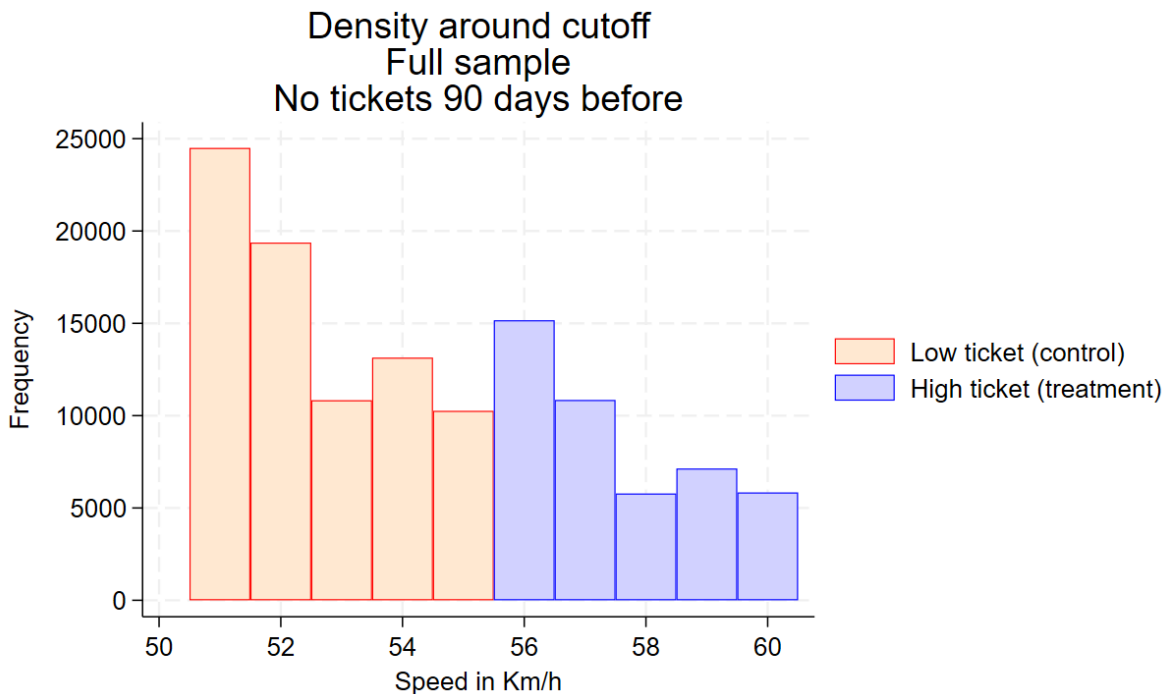


Figure 3. Frequency by condition. Full and sample.

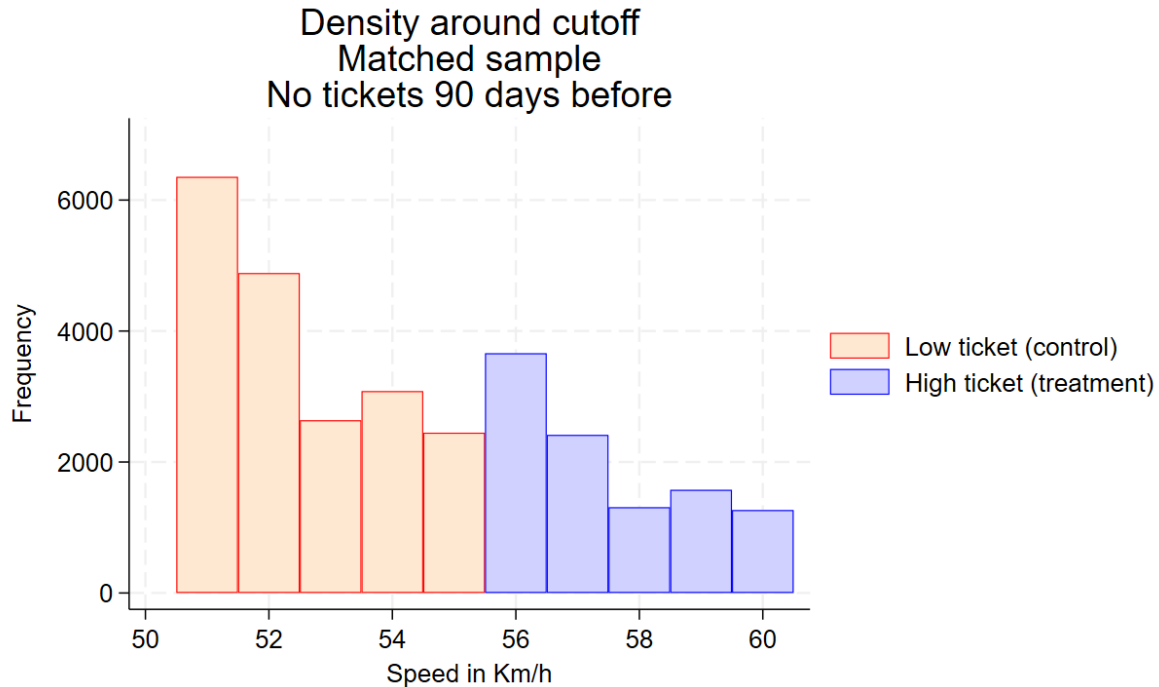


Figure 4. Frequency by condition. Matched sample.

Another way to show that drivers are not selecting themselves into not receiving a ticket is to see what happens at the posted limit. One can see that there is bunching to the left of the posted limit, but to the right of the real limit.

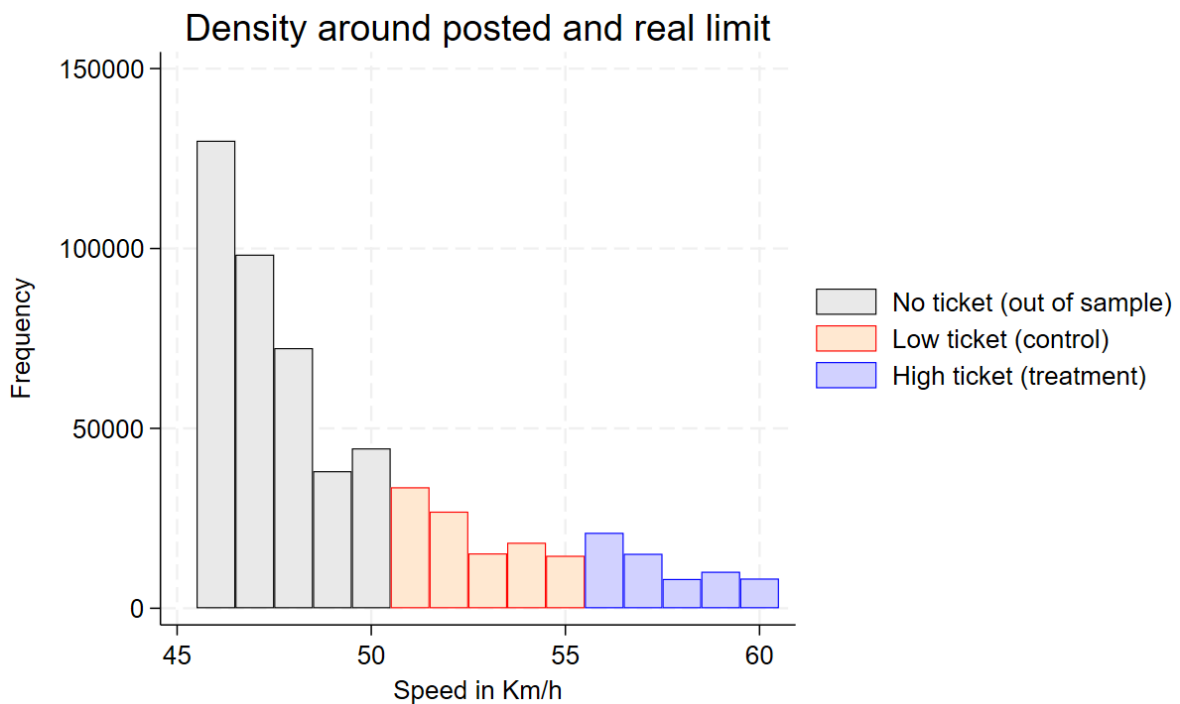


Figure 5. Discontinuity around posted and real limit.

More formal evidence of this can be given by running the manipulation test developed by (Cattaneo et al., 2018) shown in Figure 6. This is because of the discrete values of the assignment variable that does not allow for a McCrary test to be performed. Figure 4 in the Appendix perform this test with different decision regarding the polynomial degree. Figure 5 of the Appendix shows that the test is not significant in the matched sample.

More importantly, all Graphical evidence shows if there is a discontinuity it is to a higher density at the right of the cutoff. This is no evidence manipulation since in principle drivers do not want to get a ticket, so the result of these tests is unlikely to invalidate the exogeneity of the real limit.

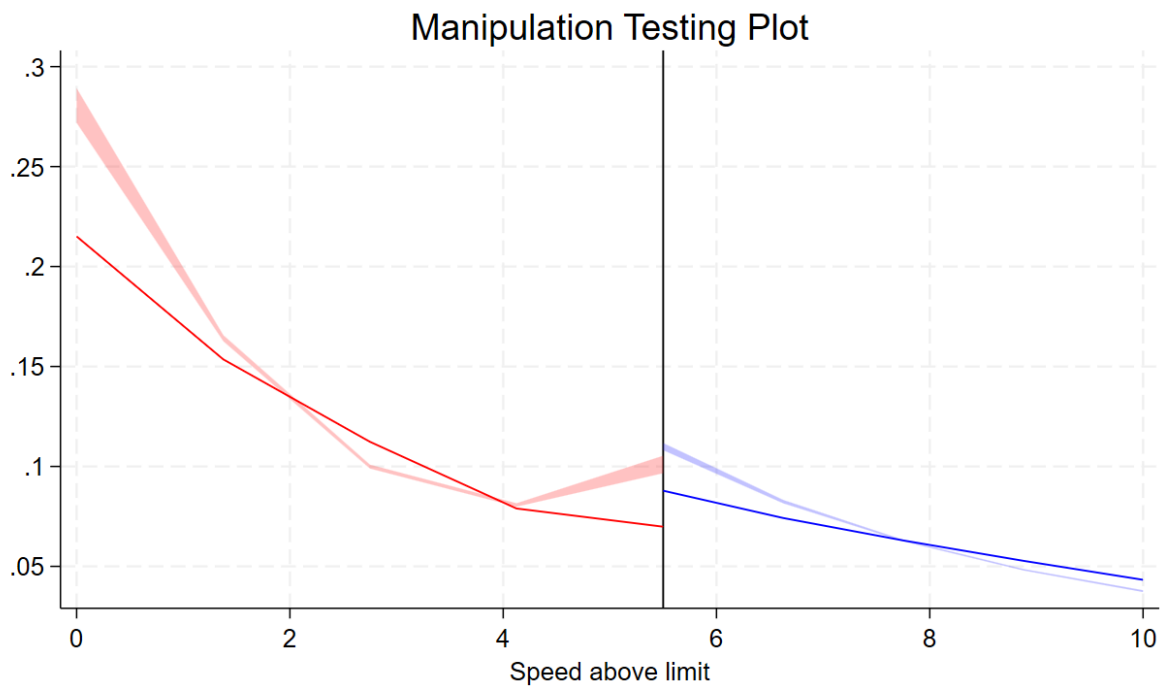


Figure 6. Manipulation test based on discontinuity. Sample with polynomial of degree 2. Full Sample.

In the analysis of the relative price, I will use the Quintiles of the distribution of vehicle price I also need to show that there is no bunching at the quintile level. Figure 7 shows this for each quintile.

Density around cutoff Matched sample, by quintile No tickets 90 days before

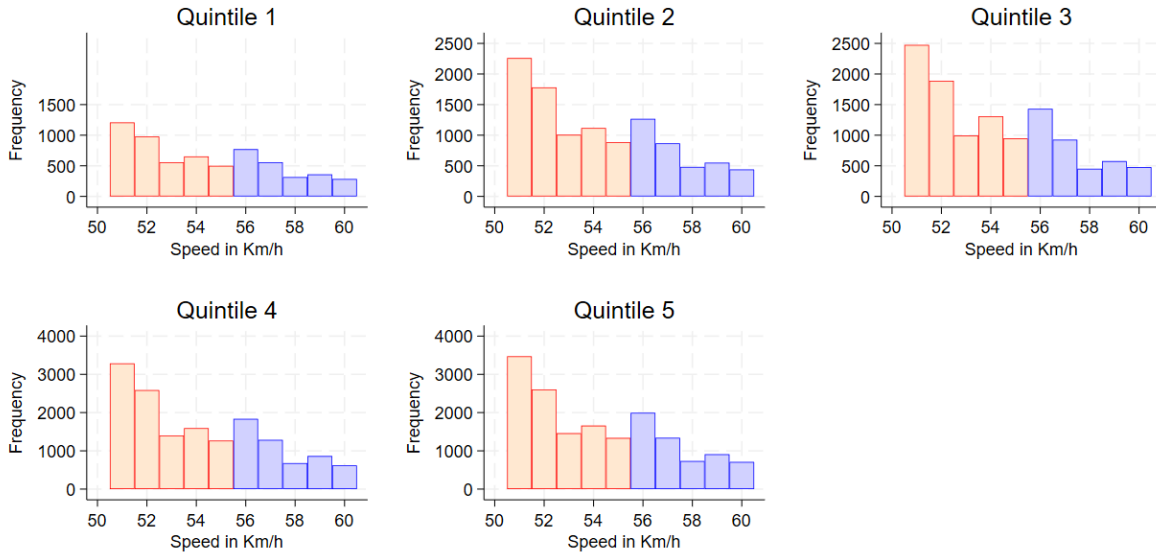


Figure 7. Frequency by condition and quintile. Matched sample.

3.3 Smoothness on observables.

3.3.1 Full sample:

Another element I need to evaluate the validity of the fuzzy regression discontinuity design is to measure the smoothness of observable variables of the vehicles involved. I will do this first on the full sample, and then on the matched sample, in which naturally I have more knowledge. For all estimations I use allow for optimal and different bandwidth below and above the cutoff.

The first variable I explore is the hour of the day at which the ticket is received, a possible indications that drivers are different in the treatment and control groups. The same selection can be analyzed with day of the week as a continuous variable or a dummy if the ticket was given during the weekend. Since in Colombia, plates for motorcycles have a different configuration than those of other vehicles¹¹, I can observe the type of vehicle involved. I use a dummy that takes the value of 1 if the vehicle involved is a motorcycle. Selection is a big concern when analyzing fines and is the reason I estimate an RD in the first place. In any case, it is important to observe that treated and not treated do not differ in their usage of the road. I compare the number of times each vehicle passes under a Speed Camera from August 2024 to February 2025. As Table 7 shows, the only difference that one can observe is by

¹¹ Motorcycle plates end in a letter, whether all other vehicle's plates end in a number. As will be seen later, for the matched sample I have much more knowledge about the vehicle.

vehicle type, and it is still small, less than 1%. In any case, in the following I will add type of vehicle as a covariate that affects the treatment status. This will be less of an issue on the matched sample, as is clear from Table 8.

Table 8: Smoothness of observables. Full sample

	(1) Hour of the day	(2) Day of the week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before
Bias-corrected	0.188 (0.248)	0.0146 (0.148)	-0.0345 (0.0338)	-0.0867*** (0.0336)	-0.569** (0.274)
Mean of not ticketed	11.80	4.156	0.350	0.347	3.682
<i>N</i>	136533	136533	136533	136533	136533
<i>N</i> left of cutoff	52468	52468	52468	52468	52468
<i>N</i> right of cutoff	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8. Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess.

3.3.2 Matched sample:

In this case the total number of available observations is 67,235¹², around 40% of the Full Sample. Just like with the full sample, most of the variables are smooth at the cutoff. It is also the case that there are fewer motorcycles are the cutoff, but one should note the significant difference in percentages between the two samples. Whereas in the Full Sample 34% of vehicles are motorcycles, this percentage is only 7% in the matched sample. Note also that the Matched Sample has a higher number of observations before the first excess, meaning that these vehicles are more frequent users of the road than the general sample. Both facts reinforce my argument about the Matched sample not being a representative part of the Full Sample.

¹² As I explained before, the vast majority of those (62,288 or 92% of the matched sample) correspond to vehicles that are registered in Bogotá. The rest are registered in other municipalities but pay an exemption for the congestion restriction that exists in the city that allow local authority access to the vehicle data.

Note that there is no significant difference in terms of observations before the speed excess and price. Both facts are also central to my estimation later on. In this case I can also evaluate the type of service which the vehicle is assigned to through a dummy that takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services). I can also say a little more about the vehicle type, besides not being a motorcycle. In this case I construct a dummy that takes the value of 1 if a vehicle is an SUV like vehicle. Both of these additional variables are smooth at the cutoff.

Table 9: Smoothness of observables Matched sample

	(1) Hour of day	(2) Day of week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before	(6) Price of vehicle	(7) Service type	(8) SUV dummy
Bias-corrected	-0.288 (0.287)	0.192 (0.178)	-0.0205 (0.0406)	-0.0581*** (0.0210)	-0.231 (0.364)	-7.562* (4.425)	-0.0202 (0.0294)	-0.0601 (0.0388)
Mean of not ticketed	11.97	4.405	0.429	0.0138	4.637	34.19	0.153	0.328
N	67235	67235	67235	67235	67235	67235	67235	67235
N left of cutoff	26548	26548	26548	26548	26548	26548	26548	26548
N right of cutoff	26909	26909	26909	26909	26909	26909	26909	26909
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9. Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess. Type of service takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services) and 0 otherwise. SUV dummy takes the value of 1 if vehicle is an SUV and 0 otherwise.

Also central to my future estimation is verifying the smoothness at each quintile level. This is shown on Tables 6 to 10 of the Appendix. In general, there is smoothness, but it increases as the quintile level goes up. Since most of my findings are only relevant at quintile 5 this is reassuring for my estimation strategy. In any case, all the observables that are significant will be used as covariables as it is done with the motorcycle dummy.

4. General results.

In this section I show all the results of the estimations. In section 4.1 I presents all results on reoffending – speeding. In Section 4.2 I presents results on Behavioral change – accidents. Chapter 5 is dedicated to the relative price analysis.

4.1 Reoffending – Speeding:

The most basic deterrent effect of a speeding ticket is to prevent speeding itself. This is why I call this behavior reoffending. Note that the assumption that drivers do not know the real speed limit implies that reoffending is measured by exceeding the posted limit. Since the risk of injury and death grows with the level of speeding it is also important to measure the deterrent effect on the most serious infractions.

4.1.1. Full sample:

Table 10 presents the results for equation (1) with different dependent variables related to speeding for a period of six month after a vehicle is observed speeding under a Camera. Moderate Speeding relates to the fact that a vehicle was observed at a speed between 50 km/h and 60 km/h at least once. Serious speeding (Speeding>10 in the table) refers to the fact that a vehicle was observed at a speed higher than 60 km/h at least once. Odd columns refer to a dummy that measures if the vehicle ever had the behavior (extensive margin), while even columns measure the total number of times each vehicle had the behavior (intensive margin). Most, but far from all, vehicles commit a speeding infraction only once.

The results point at ticket having a causal effect of reducing all types of speeding behavior. This reduction is significant in magnitude, both at the extensive and intensive margins. Very important is the fact that the result stands for serious speeding, even if its probability is a quarter of the probability of moderate speeding.

Table 10: Effect of receiving a ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.131*** (0.0278)	-0.221*** (0.0527)	-0.0651*** (0.0164)	-0.0744*** (0.0250)
Mean of not ticketed	0.195	0.298	0.0555	0.0716
<i>N</i>	136533	136533	136533	136533
N Left of cutoff	52468	52468	52468	52468
N Right of cutoff	57605	57605	57605	57605
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle was captured

in Moderate speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one Serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle was captured in Serious Speeding.

A central question for policy makers is the effectiveness of speeding tickets and other measures for speed control in the face of the growing number of motorcycles in middle income countries. As I explained before, the existing data only allows me to build a dummy that says if a vehicle is a motorcycle (=1) or not (=0). As is shown in Figure 2 of the Appendix, even if the discontinuity is smaller for motorcycles, it is still present for both values of my dummy variable. Table 11 (Not motorcycles) and 12 (Motorcycles) of the Appendix show that deterrent effect is only present for not-motorcycles.

4.1.2. Matched sample:

I can also do this analysis in the matched sample. As Table 11 shows the results are essentially the same, a general effect of reduction in all types of speeding. Note also that the result for Serious Speeding is slightly smaller in magnitude and less robust than in the general sample.

Table 11: Effect of receiving a ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.283*** (0.0361)	-0.405*** (0.0727)	-0.0521*** (0.0191)	-0.0546** (0.0265)
Mean of not ticketed	0.258	0.405	0.0565	0.0687
N	67235	67235	67235	67235
N Left of cutoff	26548	26548	26548	26548
N Right of cutoff	26909	26909	26909	26909
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11. Reoffending. Matched sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

Since my matched sample is almost exclusively comprised of not-motorcycles, in this case there is no point in doing this analysis by vehicle type.

4.2 Behavioral change – accidents:

As I argued in the introduction, speeding tickets have an indirect deterrent purpose. They are a tool for preventing injuries and deaths caused by traffic accidents. Thus the most important causal estimate for public policy is whether in fact speeding tickets prevent accidents.

Table 12 presents the results of equation (1) when the dependent variable measures accidents. Neither a dummy for having an accident (column 1), nor the continuous variable for total of accidents (column 2), nor the dummy for accident with at least one fatality¹³ (column 3) are significant. Nor is it useful to separate the sample into cars and motorcycles.

The effect on the matched sample is equally null, so I do not report it.

Table 12: Effect of receiving a ticket

	(1) Any accident dummy	(2) Total Accidents continuous	(3) Deadly accident dummy	(4) Cars dummy	(5) Motorcycles dummy
Bias-corrected	-0.00143 (0.0131)	-0.00210 (0.0166)	0.00441 (0.00359)	-0.00337 (0.0100)	0.000703 (0.0328)
Mean of not ticketed	0.0363	0.0420	0.00258	0.01317	0.08106
N	136533	136533	136533	89037	47496
N Left of cutoff	52468	52468	52468	35012	17456
N Right of cutoff	57605	57605	57605	35827	21778
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12. Behavioral change. Full sample. Column 1 (Any accident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (Total Accidents) is a continuous variable; the total of times a vehicle had an accident. Column 3 (Deadly Accident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality or Triage I attention in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles.

A possible objection to these results could be the fact that the estimation is underpowered. I should not that the same number of observations produced a very credible result for speeding. I should also point out that the number of accidents in

¹³ I also include here those accident that required immediate medical attention because of immediate risk of death, Triage I, according to the Triage categories defined by the Colombian Minister of Health. 8% of the accidents that required medical attention are considered Triage I.

the sample is small, but not negligible. There are 7,162 (108 of them serious) accidents in my estimation sample. These are around 6.2% of the total 116,088 total accidents that occurred in the period of study. In addition, I would say that 3.6% of vehicles in the control sample do have an accident, so it is a rare but not extremely rare phenomena¹⁴.

I would argue that the effect is too small to be detected, where it big my estimation would detect it.

At the risk of overinterpreting null effects, this table seems to be in line with what I have shown in the previous estimations. Speeding tickets seem to have a deterrent effect of accident in non-motorcycles that is absent in motorcycles. This might also help explain the following results.

5. Relative price.

The central questions of this paper is whether the deterrent effect of fines is independent of their relative price. Many countries have fines that are relative to income, mainly out of justice considerations. I am thinking mainly about effectiveness for preventing unwanted behavior.

Table 13 reproduces equation (1) with the dependent variable of a dummy of speeding (column 1 of table 10). The quintile corresponds to the commercial price of the vehicle as proxy for the owner's income. One can see that the effect is concentrated on the higher quintiles of the distribution. Both magnitude and significance are lower for the lower quintiles, in fact the effect is increasing on the quintile of vehicle price. As Table 13 of the Appendix shows this result is robust to using the continuous variable of the total times a vehicle is seeing under Minor Speeding, with the most negative coefficient also found in the highest quintile.

But Table 14 of the Appendix, also shows a complex picture when using Serious Speeding. In this case, the effect is not higher in the higher quintiles, but on Quintile 2. I will come back to this puzzle when discussing the effect of speeding tickets in other traffic infractions.

¹⁴ I can also add that speeders are much more accident prone than non-speeders. Using all observations in this period, speeders are twice more likely than non speeders to ever be involved in an accident.

Table 13: Effect of receiving a ticket

	(1) Quintile 1 Minor Speeding	(2) Quintile 2 Minor Speeding	(3) Quintile 3 Minor Speeding	(4) Quintile 4 Minor Speeding	(5) Quintile 5 Minor Speeding
Bias-corrected	-0.0587 (0.109)	-0.0250 (0.0707)	-0.109 (0.0843)	-0.208*** (0.0645)	-0.178** (0.0721)
Mean of not ticketed	0.145	0.211	0.216	0.234	0.243
<i>N</i>	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
<i>N</i> Left of cutoff	2692	4794	5141	6854	7067
<i>N</i> Right of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13. Reoffending by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one speeding between 50 and 60km/h, and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

Table 14 does the same for the dummy of ever being in an accident. A similar finding is seen here. The desired effect is increasing in the vehicle quintile, and only negative for the higher quintiles. Quintile 5 is the only significant and negative result for accidents, while the result is significant but in the undesired direction on Quintile 2. Note that the rates of accidents are different than those for speeding. While speeding is more common on higher quintiles, accidents are concentrated on the lower ones. Thus, while speeding tickets are well targeted at reducing speeding, but they are poorly targeted at preventing accidents.

Table 15 of the Appendix shows a much more nuanced finding when talking about serious accidents. In this case coefficient is positive in most quintiles but is of a higher magnitude at quintile 1 than any other, in fact it still decreasing on quintile level. I would advocate for not overinterpreting results on serious accidents, since they are extremely rare.

Table 14: Effect of receiving a ticket

	(1) Quintile 1 Any accident	(2) Quintile 2 Any accident	(3) Quintile 3 Any accident	(4) Quintile 4 Any accident	(5) Quintile 5 Any accident
Bias-corrected	0.0307 (0.0612)	0.0503*** (0.0164)	0.0335 (0.0223)	-0.0247 (0.0193)	-0.0661*** (0.0239)
Mean of not ticketed	0.0443	0.0121	0.0113	0.0142	0.0177
<i>N</i>	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Right of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14. Behavioral change by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one accident, and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

All this section points in the same direction, relative price does matter and in the case of Bogotá it seems to suggest that tickets are more effective not when they are more, but less expensive. A natural question for these results is how much they are the effect of taxis and other public service vehicles. I can take them out of the sample and all results (not reported) are in line with the story told in this chapter.

6. Robustness checks.

In this section I introduce some further evidence that shows my results are credible. First, I present alternative estimations to show consistency. Second, I use an alternative method of estimating my results: an IV-judge type of estimation using data on the police agent in charge of verifying each speeding ticket.

6.1 Robustness to some decisions:

Figure 15 shows a comparison between the coefficient found on column 1 of Table 10 and the coefficient found with different decisions. In particular, whether to include vehicles that had a previous fine¹⁵, to use vehicles that pass more than once under a Camera and to have a bigger time for selecting my sample.

The most important choice is to include vehicles that have more than one excess in my sample period. There are at least two alternatives for including these vehicles. Since the problem is really how to assign treatment, my choice is to consider vehicles

¹⁵ As I argued, I want to separate the effect of this fine from all previous fines. Including those vehicles that had a fine in the previous six months increases the sample to 145145 (from 136553), a 6.2% increase.

that passed always below or above the real limit and use their average speed and first observation as treatment time (Repeated observations, one side on Figure 15). I find this better to the convention of using the first time the vehicle exceeds as its treatment assignment. Since there is a delay between speeding and the moment in which a vehicle is notified of the infraction, a vehicle can be treated and not know it for many days (Repeated observations, first on Figure 15).

The coefficient is very stable across all decisions. In fact, including previous observations only increases the magnitude of the effect.

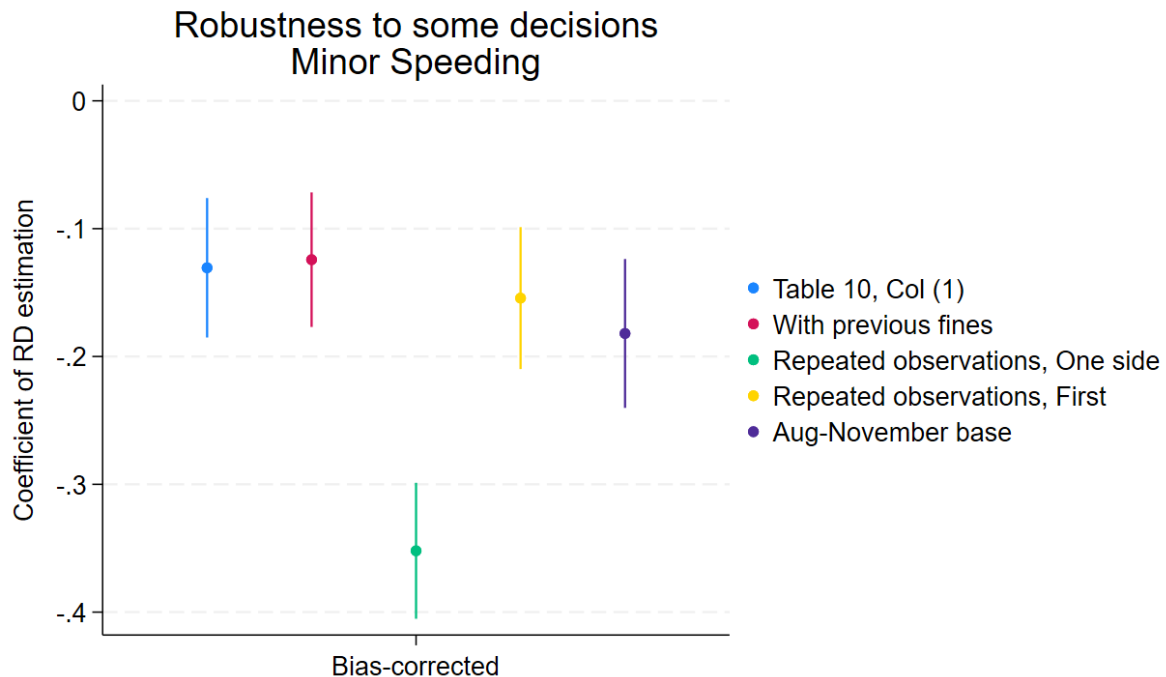


Figure 15. “Use previous fines” uses the same estimation, this time including vehicles that have had a ticket in the previous six months. “Repeated observations, One Side” shows the coefficient using Vehicles that passed more than once and always above or below the cutoff. “Repeated observations, First” shows the coefficient using Vehicles that passed more than once using the first observations as treatment assignment. “Aug-November” base makes the database bigger, by including all vehicles from August 16 to November 15, 2024.

Figure 16 and 17 of the Appendix present the same calculations for other dependent variables: Serious Speeding and Any accident. The coefficients are very stable to most of the decisions I make in the general case, but a longer period of study changes some of the results. I should point that smoothness of the variables is less strong on those bigger samples, the reason I do not use them in the first place.

6.2 An IV-judge using police officer manual validation:

According to Colombian law all speeding tickets must be subject to human verification. This has to do mainly with the fact that the technology for taking pictures

of vehicles while speeding is not perfect, and some pictures are blurry. This task is done by trained police officers that must evaluate whether a specific picture should transform into a ticket. The pictures assigned to each policemen are random, and they have no way to refuse their verification.

I have data for the identification of all pictures of vehicles that were circulating above 56 km/h during the months of October and November. I also can see which agent was in charge of reviewing each picture. As Figure 16 shows, there is considerable heterogeneity in the percentage of pictures that each agent transforms into a ticket.

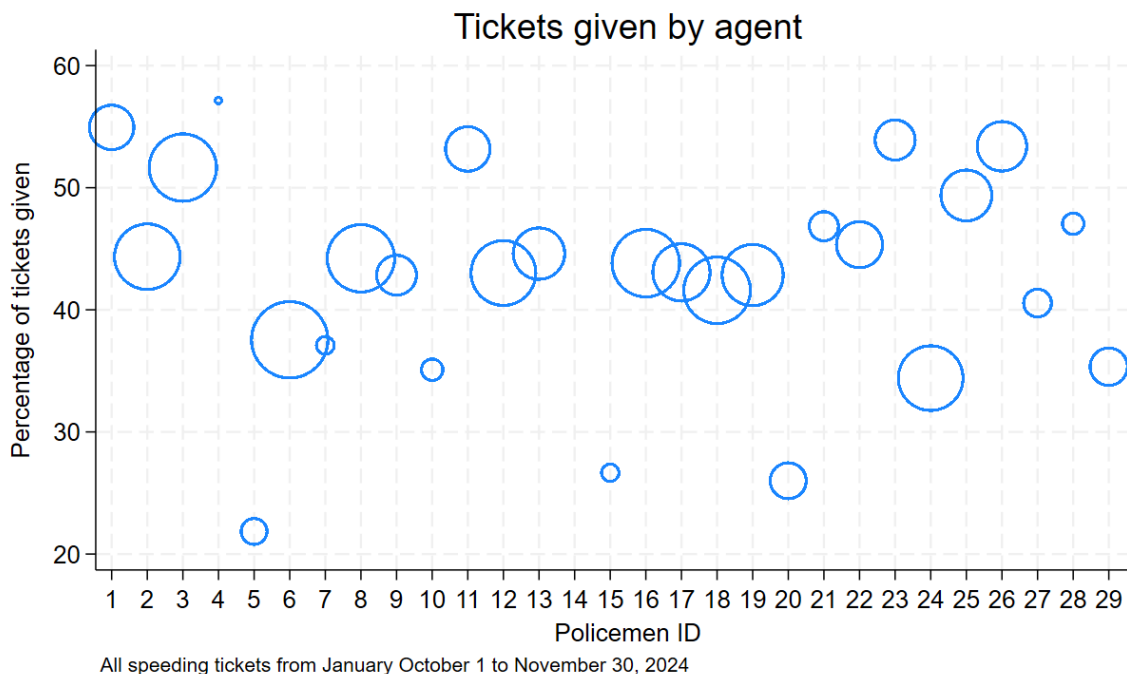


Figure 16. Policemen ID is a number to identify each police officer. The size of the circle is proportional to the number of pictures reviewed by each agent.

Figure 16 indicates that there might be some randomness associated to the Policemen in charge. For this to transform into a credible instrumental variable one would want to verify that these different rates are not related to any observable characteristic of the ticket, or the vehicle involved. Figure 17 shows that the strictness is stable for all Policemen across at least two meaningful variables, whether a vehicle is a motorcycle and those vehicles that drive faster than 60 km/h.

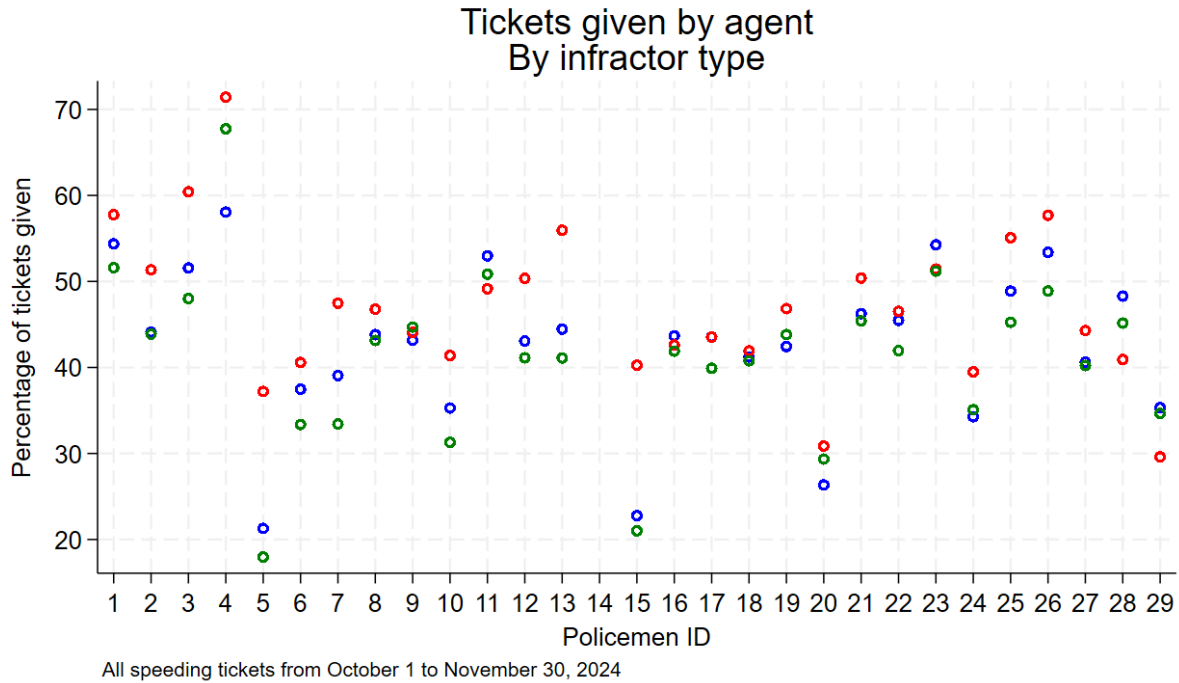


Figure 17. Policeman ID is a number to identify each police officer. Blue dots correspond to all pictures, red dots to pictures of motorcycles and green to vehicles under serious speeding (40% of this subsample).

Both previous Figures give evidence to use the random assignment of the Policemen in charge of reviewing each picture as an instrument of the ticket. I will then capture the part of receiving a ticket that is due to the randomness of the assigned policemen. I also must emphasize that this is a different sample, consisting mostly of drivers that decide to exceed speeding limits seriously. I should also note that in this case I am not finding a local effect, but the general effect on those that decide to exceed the limit by a significant amount of kilometers.

Table 18 shows the result of an estimation with the same dependent variables as Table 8. As one can clearly see, the significance and sign of all the estimation is similar, but magnitudes are considerably smaller. While the effect of a speeding ticket was a 64% reduction in the change of reoffending, in this case the effect is only 25%. This surely is caused by these drivers deciding to disobey speeding limits by a bigger amount than the other sample.

Table 18: Effect of agent reviewing ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0416*** (0.0158)	-0.0840*** (0.0279)	-0.0201** (0.00897)	-0.0257** (0.0127)
Mean of not ticketed	0.164	0.241	0.0470	0.0604
<i>N</i>	132668	132668	132668	132668
First stage Wald-F	82.56	82.56	82.56	82.56

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 18. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one Serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding. This table is comparable to Table 10.

The results on Table 18 are in line with the main finding of this paper. One could also use this sample to verify the second finding: effects being explained only by cars, not motorcycles. Table 18 and Table 19 of the Appendix do this. Note that in this case the effect is more precisely estimated for cars than motorcycles, but there is some for the latter. Once again, this does not cast doubt on previous findings but shows that serious speeders are more similar between them, independent of the type of vehicle they drive.

One could use this estimation strategy to verify the other two findings. As Table 19 shows there is also a null effect on all dependent variables related to the chance of being involved in an accident. In this case the coefficients are positive, not like those found in my RD estimation that were negative but not significant.

Table 19: Effect of agent reviewing ticket

	(1) Any accident dummy	(2) Total Accidents continuous	(3) Deadly accidents dummy	(4) Cars dummy	(5) Motorcycles dummy
Ticket	0.00484 (0.00664)	0.00837 (0.00816)	-0.0000405 (0.00182)	0.00118 (0.00463)	0.00267 (0.0182)
Mean of not ticketed	0.0246	0.0280	0.00194	0.0103	0.0630
N	132668	132668	132668	94536	38132
First stage Wald-F	82.56	82.56	82.56	65.40	28.93

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 19. Behavioral change. Full sample. Column 1 (Any accident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (Total Accidents) is a continuous variable; the total of times a vehicle had an accident. Column 3 (Deadly Accident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles. This table is comparable to Table 14.

The final finding one could verify has to do with the heterogeneous effect by vehicle price. As Table 20 shows, the effect of a ticket on speeding behavior is in this case also concentrated on the higher quintiles. Even more, like in previous cases for quintile 2 there is a positive coefficient, which is slightly not significant at conventional levels. Table 20 of the Appendix does the same with any accident as dependent variable, where results are more blurry, in part because of the smaller frequency of the studied behavior.

Table 20: Effect of agent reviewing ticket. Minor Speeding. By quintile

	(1) Matched Sample	(2) Quintile 1	(3) Quintile 2	(4) Quintile 3	(5) Quintile 4	(6) Quintile 5
Ticket	-0.0673*** (0.0226)	-0.0339 (0.0670)	0.0811 (0.0495)	-0.0868** (0.0417)	-0.102** (0.0447)	-0.107*** (0.0439)
Mean of not ticketed	0.200	0.119	0.179	0.195	0.214	0.227
N	48230	3562	8772	10223	12150	13523
First stage Wald-F	46.28	3.551	8.688	13.11	12.64	13.37

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 20. Reoffending by price quintile. Matched sample. Dependent variable is a dummy that takes the value of 1 Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise.

In summary, although the sample for the previous sections is different from this one, nor are the estimation strategies capturing the same causal effect, one can safely say that these results give more credibility to the causal effects found. All evidence

points in the same direction of tickets having a deterrent effect, and this effect being bigger for higher quintiles of vehicle prices.

7. Wake-up call or information? An exploration on mechanisms.

Speeding tickets are meant to prevent the most serious consequences of traffic accidents. In fact, speeding Cameras in Bogotá were originally called “Camaras Salvavidas” (*Lifesaving cameras*). As we have seen, in practice Cameras seem very effective at reducing speeding itself but show little effect on reducing accidents. In this section I use other behaviors to try to understand how much of the effect of traffic tickets corresponds to changes in the driver’s behavior and how local, both in geographical and chronological sense, the effect is.

The first question I want to explore is whether the results are caused by a reduction in the frequency of driving. This is not good or bad in itself in terms of road safety, but it might have significant costs at the personal level. I use as a dependent variable the total number of times a vehicle passes under a Speed Camera after the speeding infraction. Column 1 of Figure 21 shows that speeding tickets seem not to have an effect on road usage, and that this mechanism is surely not causing the results seen before.

A second question I can address is how local the effect is. For this purpose I can use the location of each Camera. Note that due to the nature of traffic management in the city of Bogotá all Cameras are not active during the six months following an speeding infraction. Instead of measuring each Camera I use the division of Bogotá into 20 Localidades (something close to Boroughs). In Columns 2 and 3 of Figure 21 I separate further speeding into close or far from the Camera at which a vehicle was captured speeding. I consider close any camera that is inside two levels of proximity, and far those occurring outside this range. Result is robust to using just one level of proximity.

A third question, and the most interesting one in behavioral terms, is whether in general speeding fines are causing a change in driving behavior or only the need for being generally more aware of the locations and speed limit of Speed Cameras. This question can be addressed by looking at other behaviors on the road that are also punished by road authorities. This is complicated by the fact that speeding is by far the most common infraction in the period of study, even if this period coincides with a city wide campaign against parking in prohibited spaces. Column 4 and 5 of Figure 21 explore both the extensive and intensive margin of this dimension. It is reassuring to see that this happens, since it suggests a broad behavioral change.

Table 21 of the Appendix examines how the result seen on Column 4 changes by quintile. There is an interesting finding since, contrary to all other effects, the most desired outcome (a big reduction in other tickets) is concentrated on quintiles 2 and 3. A tentative explanation for this finding has to do with the fact that Speeding tickets have no interaction with a policeman, nor the risk of confiscation of the vehicle,

whereas other tickets have this risk. So a speeding ticket might be a sign that the authority is present, even if it is incapable of collecting the money from speeding fines. This hypothesis of the risk of engaging in person with the police goes in line with the complex effect of tickets on Serious Speeding, another behavior that might alert human policemen.

Table 21: Behavioral mechanisms Full sample

	(1) Future observations	(2) Minor Speeding nearby	(3) Minor Speeding far	(4) Other traffic tickets dummy	(5) Other traffic tickets continuous
Bias-corrected	2.359 (1.963)	-0.105*** (0.0267)	-0.0691*** (0.0156)	-0.0867*** (0.0228)	-0.125*** (0.0387)
Mean of not ticketed	21.013	0.176	0.0504	0.117	0.160
N	136533	136533	136533	136533	136533
N Left of cutoff	52468	52468	52468	52468	52468
N Right of cutoff	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21. Behavioral change. Full Sample. Column 1 is a continuous variable that measures the total number of observations after the first speeding of a vehicle. Column 2 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h) inside a range of two proximate *Localidades* and 0 otherwise. Column 3 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h) outside a range of two proximate *Localidades* and 0 otherwise. Column 4 is a dummy variable that takes a value of 1 if the vehicle ever had another ticket for any cause after the first speeding instance, and 0 otherwise. Column 5 is a continuous variable that measures the total number of ticket for any cause after the first speeding instance.

There is also a final question about persistence of the effect of receiving a ticket in speeding. Table 22 shows that the effect is not particularly concentrated on the first months, but it seems to fade out after six months. One can say that it is reassuring that there is a significant effect after five months. Table 22 of the Appendix does the same for Serious speeding. Once again, in this case the story is much more blurry. But with some confidence one can say that the effect is present for more than three months after the day of the ticket.

Table 22: Effect of receiving a ticket. By month

	(1) Minor Speeding any time	(2) Minor Speeding Month 1	(3) Minor Speeding Month 2	(4) Minor Speeding Month 3	(5) Minor Speeding Month 4	(6) Minor Speeding Month 5	(7) Minor Speeding Month 6
Bias-corrected	-0.175*** (0.0294)	-0.0216* (0.0127)	-0.0811*** (0.0158)	-0.0139 (0.0153)	-0.0481*** (0.0153)	-0.0466*** (0.0139)	0.0187* (0.00971)
Mean of not ticketed	0.222	0.0352	0.0539	0.0488	0.0481	0.0382	0.0194
N	136533	136533	136533	136533	136533	136533	136533
N left of cutoff	52468	52468	52468	52468	52468	52468	52468
N right of cutoff	57605	57605	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 22. Behavioral change. Full Sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h) any time after the first speeding instance and 0 otherwise. Columns 2 to 7 measure the same variable by month.

8. Discussion and possible policy recommendations.

The most robust finding of this paper can be summarized in one sentence: speeding tickets causally reduce moderate speeding but seem to have a much smaller (probably null) effect in reducing accidents. As I argued, technically I have not enough statistical power to say that there is no effect on accidents, but I can confidently say that the effect is negative and statistically and practically significant for speeding. The reduction of speeding is concentrated both in automoviles and lesser speeding infractions, but at the general level there seems to be an effect on more serious speeding. This main finding is true both for the Full sample and for the Matched sample of vehicles that are registered in Bogotá.

The most interesting finding of this paper can also be summarized in one sentence: speeding tickets are more effective at deterring unwanted behaviors for those for whose they are less expensive. This comes from the fact that for both speeding and accidents, the higher the quintile of vehicle price the most deterrent speeding tickets are. In fact, all of the desired outcomes occur at quintile 5 of the vehicle price distribution and most of the undesired at quintile 2. Since accident are concentrated on the lower quintiles, speeding tickets seem to be a specially bad targeted intervention.

Gary Becker wrote that “Fines are preferable to imprisonment and other types of punishment because they are more efficient” (Becker, 1992). In his analysis “the social cost of fines is about zero”, and the optimal amount of a fine “depend[s] only on the marginal harm and cost and not at all on the economic positions of offenders”(Becker, 1974). To be fair, Becker was surely also thinking of more serious crimes that a speed of 56 km/h, but those strong statements are also a clear representation of his thinking.

This paper shows that fines can be socially inefficient in a very particular way. Speeding fines are just one example of many where the State punishes a proxy behavior (speeding) because it cannot punish the most undesired outcome (a car accident). I think my evidence is strong in the direction that fines are not serving their main purpose. A fine that does not deter is not socially efficient. On top of that, it seems clear in the context I study relative price does matter, and is an additional source of inefficiency since cheaper fines might reduce negative road behavior and increase the amount of money received by the government. This last idea can be supported with existing data on ticket payment, shown in Figures 23 and 24 of the Appendix.

The second inefficiency might be particularly relevant for thinking about enforcement on developing countries. Becker's logic might need to be complicated in contexts where the State can impose a fine but it can do little to enforce it. As I explained before, in the context of Bogotá the only real effect of not paying a speeding fine is the exclusion from the formal market, both in the market for used vehicles and the formal labor market due to the risk of getting a wage garnishment for unpaid fines. Fixed monetary fines might turn into an additional barrier to formality for the poorest members of society.

All this points to a clear policy recommendation: income based fines. Seeding tickets might have a better effect if they were adjusted, to a lower value, for the less wealthy drivers. If one adds the known fact that higher quintiles have higher payment rates, one can conclude that the State will prevent more speeding, some accidents, and even collect more resources if fines are lower for the poorer. At the same time this might reduce the barrier to formality that existing fines can be.

I would argue that the main effect for public policy of the findings is that the presumed causal relation between speed and accidents is less important than assumed. If speeding tickets have a significant and negative effect on speeding behavior but they do not have any significant effect on accidents (even in one case an undesired effect), then speed is not a significant cause for accidents. This has a clear policy application: other measures are needed in order to prevent road accidents from becoming an even bigger public health challenge.

The fact that the desired effects of the policy are concentrated on automobiles, and absent in motorcycles also has important policy implications. This might mean that speeding tickets are not the right tool for the changing face of the public health challenge of road fatalities that some cities are facing, since motorcycle drivers are a growing percentage of road fatalities both as victims and perpetrators¹⁶. Other interventions are needed if a city like Bogotá (and other in middle income countries) want to face the public health challenge of motorcycles (in all its dimensions:

¹⁶ In Bogotá, motorcycle users were present in 45.5% of accidents with a death in 2015. This number has steadily grown to 68.5% in 2024.

(Martínez-González et al., 2022)). Both non-automated control and behavioral interventions are urgently needed.

I don't think that an implication of the evidence shown is that speed cameras don't work. What this paper has shown is that they do not work through a very specific channel: the behavioral change of marginal speeders just above an unknown real fine threshold. Most evaluations seem to show that in the aggregate cameras work, and this might be easily explained by their effect on regular speed-limit obeying drivers both internationally (Ang et al., 2020) and specifically in Bogotá (Ferro, 2024). Newer policies might be needed for the selected population that knowing the speed limit and how strongly it is enforced, decides to speed. For the high majority of drivers knowing that there is police enforcement of speeding laws is sufficient to positively influence their behavior.

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Appendix:

1	158,703	87.44	87.44
2	17,731	9.77	97.21
3	3,460	1.91	99.12
4	934	0.51	99.64
5	371	0.20	99.84
6	142	0.08	99.92
7	52	0.03	99.95
8	39	0.02	99.97
9	21	0.01	99.98
10	10	0.01	99.99
11	13	0.01	99.99
12	3	0.00	99.99
13	6	0.00	100.00
15	1	0.00	100.00
17	1	0.00	100.00
18	1	0.00	100.00
26	1	0.00	100.00
Total	181,489	100.00	

Table 1. Times each plate passed exceeding the limit under the Cameras.

Percentage of vehicles that receive a speeding ticket
by vehicle type
Full sample

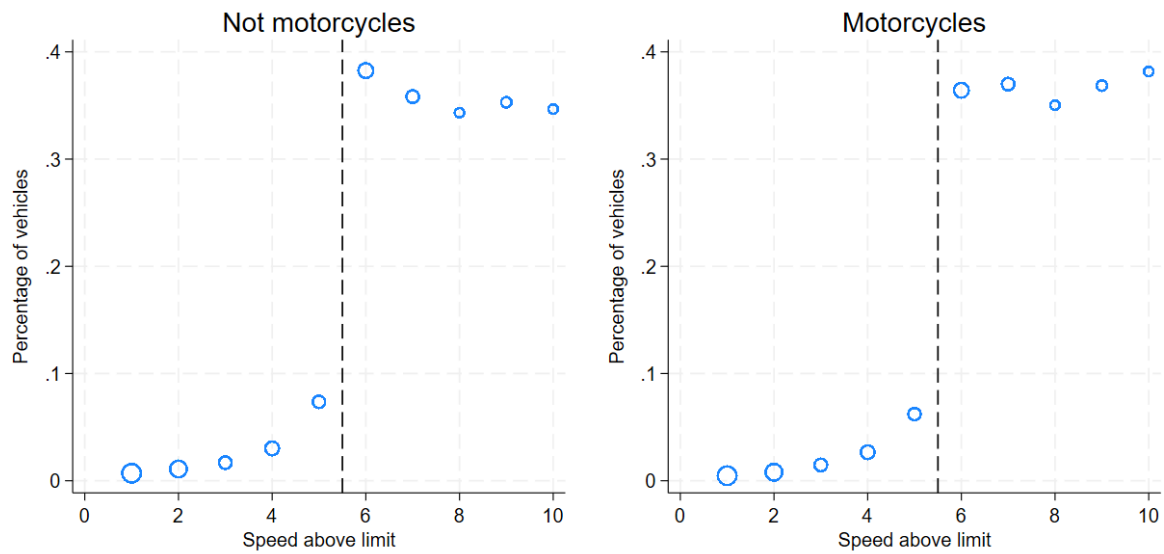


Figure 2. Existence of the discontinuity by type of vehicle. Full Sample.

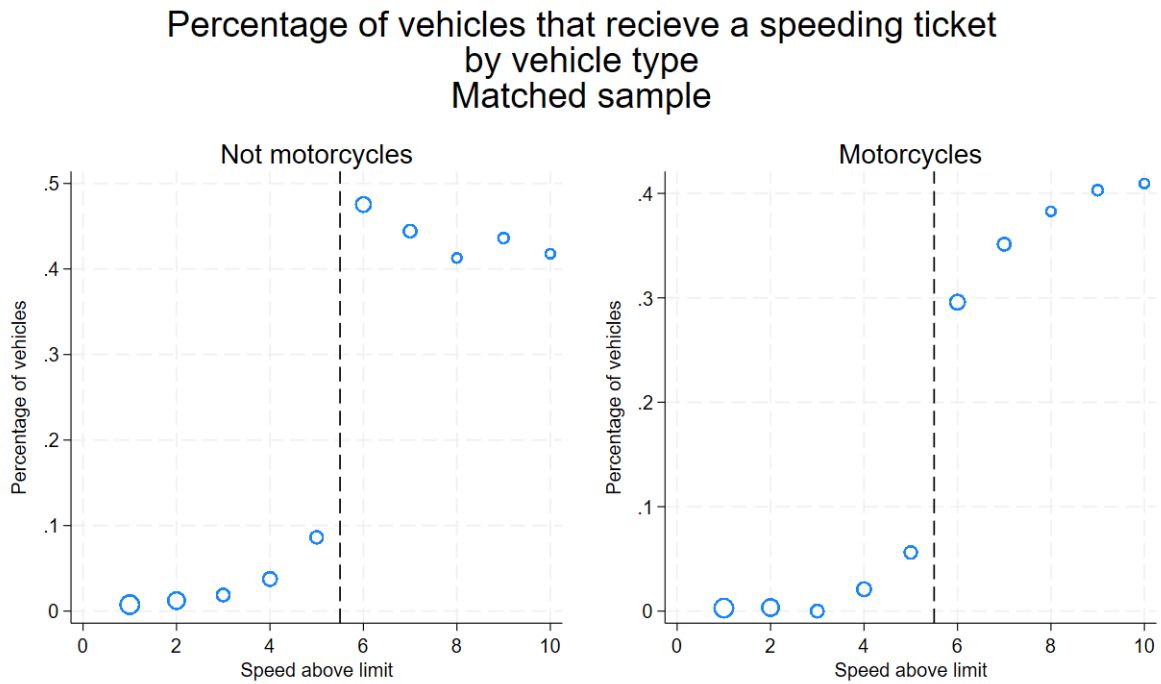


Figure 3. Existence of the discontinuity by type of vehicle. Matched Sample.

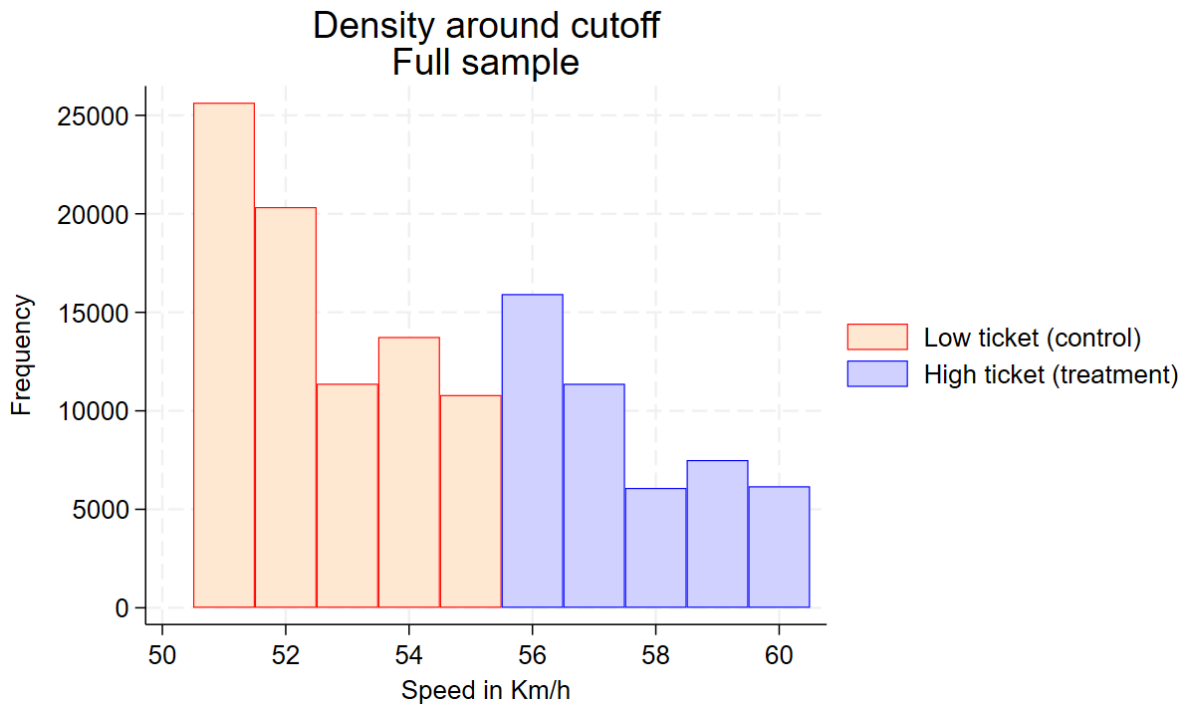


Figure 4. Density around cutoff. Matched sample.

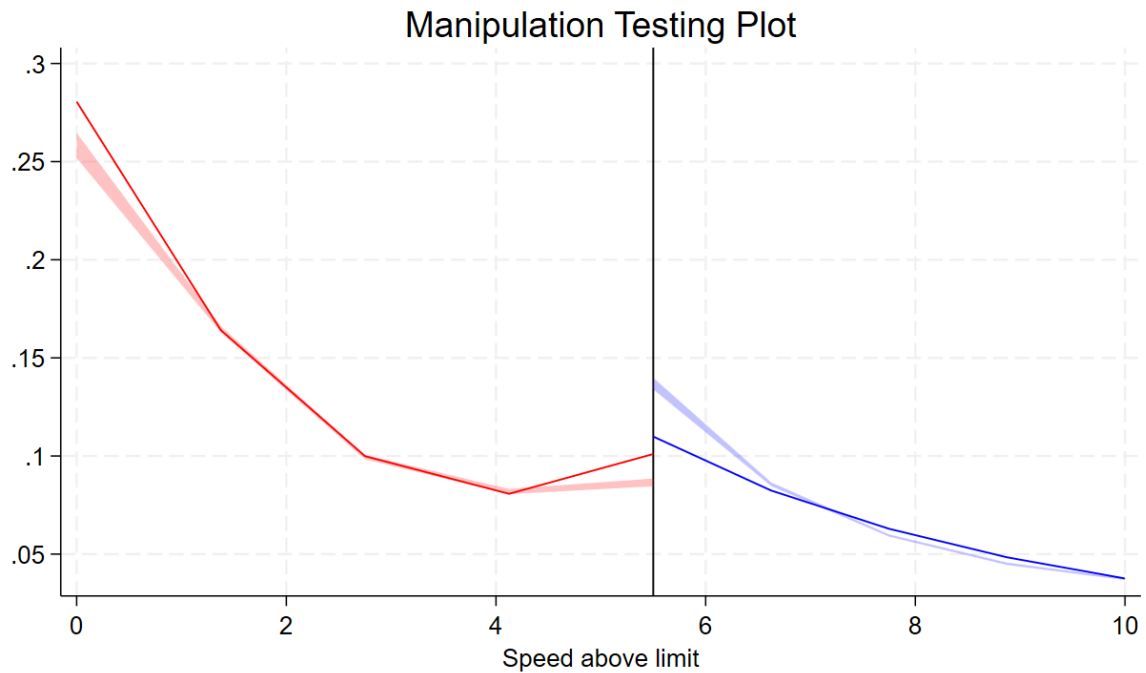


Figure 4. Manipulation test based on discontinuity. Sample with polynomial of degree 3.



Figure 5. Manipulation test based on discontinuity. Sample with polynomial of degree 2. Matched Sample.

Table 6: Smoothness of observables Matched sample. Quintile 1

	(1) Hour of day	(2) Day of week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before	(6) Price of vehicle	(7) Service type	(8) SUV dummy
Bias-corrected	-0.681 (1.035)	1.671*** (0.612)	0.439*** (0.141)	-0.699*** (0.147)	-1.078 (0.723)	0.533 (1.539)	0.0365 (0.0357)	0.00629 (0.0755)
Mean of not ticketed	11.69	4.219	0.367	0.528	2.304	9.211	0.0237	0.0747
<i>N</i>	7065	7065	7065	7065	7065	7065	7065	7065
<i>N</i> left of cutoff	2692	2692	2692	2692	2692	2692	2692	2692
<i>N</i> right of cutoff	3046	3046	3046	3046	3046	3046	3046	3046
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6. Smoothness on observables Quintile 1. Matched sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess.

Table 7: Smoothness of observables Matched sample. Quintile 2

	(1) Hour of day	(2) Day of week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before	(6) Price of vehicle	(7) Service type	(8) SUV dummy
Bias-corrected	-1.187* (0.615)	0.0316 (0.379)	-0.0431 (0.0861)	0.119*** (0.0355)	-1.501** (0.634)	-1.412*** (0.504)	-0.130** (0.0569)	0.0216 (0.0459)
Mean of not ticketed	11.87	4.367	0.415	0.0430	4.478	23.98	0.122	0.0792
<i>N</i>	11848	11848	11848	11848	11848	11848	11848	11848
<i>N</i> left of cutoff	4794	4794	4794	4794	4794	4794	4794	4794
<i>N</i> right of cutoff	4603	4603	4603	4603	4603	4603	4603	4603
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7. Smoothness on observables Quintile 2. Matched sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess.

Table 8: Smoothness of observables Matched sample. Quintile 3

	(1) Hour of day	(2) Day of week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before	(6) Price of vehicle	(7) Service type	(8) SUV dummy
Bias-corrected	-0.832 (0.727)	-0.237 (0.456)	-0.231** (0.104)	-0.0391* (0.0233)	2.165*** (0.730)	0.469 (0.688)	-0.196*** (0.0695)	0.0512 (0.0762)
Mean of not ticketed	11.97	4.405	0.429	0.0138	4.637	34.19	0.127	0.174
<i>N</i>	12824	12824	12824	12824	12824	12824	12824	12824
<i>N</i> left of cutoff	5141	5141	5141	5141	5141	5141	5141	5141
<i>N</i> right of cutoff	5010	5010	5010	5010	5010	5010	5010	5010
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8. Smoothness on observables Quintile 3. Matched sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess.

Table 9: Smoothness of observables Matched sample. Quintile 4

	(1) Hour of day	(2) Day of week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before	(6) Price of vehicle	(7) Service type	(8) SUV dummy
Bias-corrected	-0.213 (0.516)	0.00930 (0.325)	-0.0901 (0.0739)	-0.0455** (0.0177)	0.398 (0.687)	0.406 (1.151)	0.0806 (0.0585)	-0.0410 (0.0730)
Mean of not ticketed	11.97	4.352	0.408	0.0114	5.003	51.31	0.190	0.386
<i>N</i>	17225	17225	17225	17225	17225	17225	17225	17225
<i>N</i> left of cutoff	6854	6854	6854	6854	6854	6854	6854	6854
<i>N</i> right of cutoff	6820	6820	6820	6820	6820	6820	6820	6820
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9. Smoothness on observables Quintile 4. Matched sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess.

Table 10: Smoothness of observables Matched sample. Quintile 5

	(1) Hour of day	(2) Day of week	(3) Weekend dummy	(4) Motorcycle dummy	(5) Observations per day before	(6) Price of vehicle	(7) Service type	(8) SUV dummy
Bias-corrected	0.939 (0.576)	0.214 (0.357)	0.0281 (0.0809)	0.0000765 (0.00907)	-1.087 (0.941)	-12.59 (11.84)	0.0947 (0.0667)	-0.121 (0.0793)
Mean of not ticketed	12.07	4.236	0.380	0.00356	5.727	119.1	0.206	0.647
<i>N</i>	18273	18273	18273	18273	18273	18273	18273	18273
N left of cutoff	7067	7067	7067	7067	7067	7067	7067	7067
N right of cutoff	7430	7430	7430	7430	7430	7430	7430	7430
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10. Smoothness on observables Quintile 5. Matched sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess.

Table 11: Effect of receiving a ticket. Not motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.186*** (0.0365)	-0.290*** (0.0707)	-0.0814*** (0.0204)	-0.0923*** (0.0314)
Mean of not ticketed	0.221	0.344	0.0571	0.0736
<i>N</i>	89037	89037	89037	89037
N Left of cutoff	35012	35012	35012	35012
N Right of cutoff	35827	35827	35827	35827
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11. Reoffending. Not motorcycles, full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise.

Column 4 is a continuous variable, the total of times a vehicle was captured in Serious Speeding.

Table 12: Effect of receiving a ticket. Motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.0246 (0.0417)	-0.0827 (0.0737)	-0.0341 (0.0274)	-0.0407 (0.0412)
Mean of not ticketed	0.145	0.206	0.0524	0.0678
<i>N</i>	47496	47496	47496	47496
N Left of cutoff	17456	17456	17456	17456
N Right of cutoff	21778	21778	21778	21778
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12. Reoffending. Motorcycles, full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Serious Speeding.

Table 13: Effect of receiving a ticket

	(1) Quintile 1 Minor Speeding Continuous	(2) Quintile 2 Minor Speeding Continuous	(3) Quintile 3 Minor Speeding Continuous	(4) Quintile 4 Minor Speeding Continuous	(5) Quintile 5 Minor Speeding Continuous
Bias-corrected	-0.214 (0.184)	0.156 (0.119)	0.190 (0.137)	-0.340*** (0.123)	-0.469*** (0.135)
Mean of not ticketed	0.199	0.308	0.317	0.344	0.357
<i>N</i>	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Right of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13. Reoffending by quintile of vehicle price. Matched sample. Dependent variable is a continuous variable that captures the total times a vehicles is observed between 50 and 60km/h after the first one. For each quintile, variables that are not smooth at cutoff are used as covariates.

Table 14: Effect of receiving a ticket

	(1) Quintile 1 Serious Speeding Dummy	(2) Quintile 2 Serious Speeding Dummy	(3) Quintile 3 Serious Speeding Dummy	(4) Quintile 4 Serious Speeding Dummy	(5) Quintile 5 Serious Speeding Dummy
Bias-corrected	0.0263 (0.0635)	-0.0841** (0.0391)	-0.0478 (0.0456)	-0.0563* (0.0341)	-0.0343 (0.0415)
Mean of not ticketed	0.0429	0.0519	0.0513	0.0540	0.0690
<i>N</i>	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
<i>N</i> Left of cutoff	2692	4794	5141	6854	7067
<i>N</i> Right of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14. Reoffending by quintile of vehicle price. Matched sample. is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

Table 15: Effect of receiving a ticket

	(1) Quintile 1 Any serious accident	(2) Quintile 2 Any serious accident	(3) Quintile 3 Any serious accident	(4) Quintile 4 Any serious accident	(5) Quintile 5 Any serious accident
Bias-corrected	0.0556*** (0.0172)	0.00557 (0.00559)	0.0216*** (0.00423)	-0.00271 (0.00171)	0.0141*** (0.00527)
Mean of not ticketed	0.00326	0.000928	0.00117	0.000813	0.00131
<i>N</i>	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
<i>N</i> Left of cutoff	2692	4794	5141	6854	7067
<i>N</i> Right of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15. Behavioral change by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one serious accident, and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

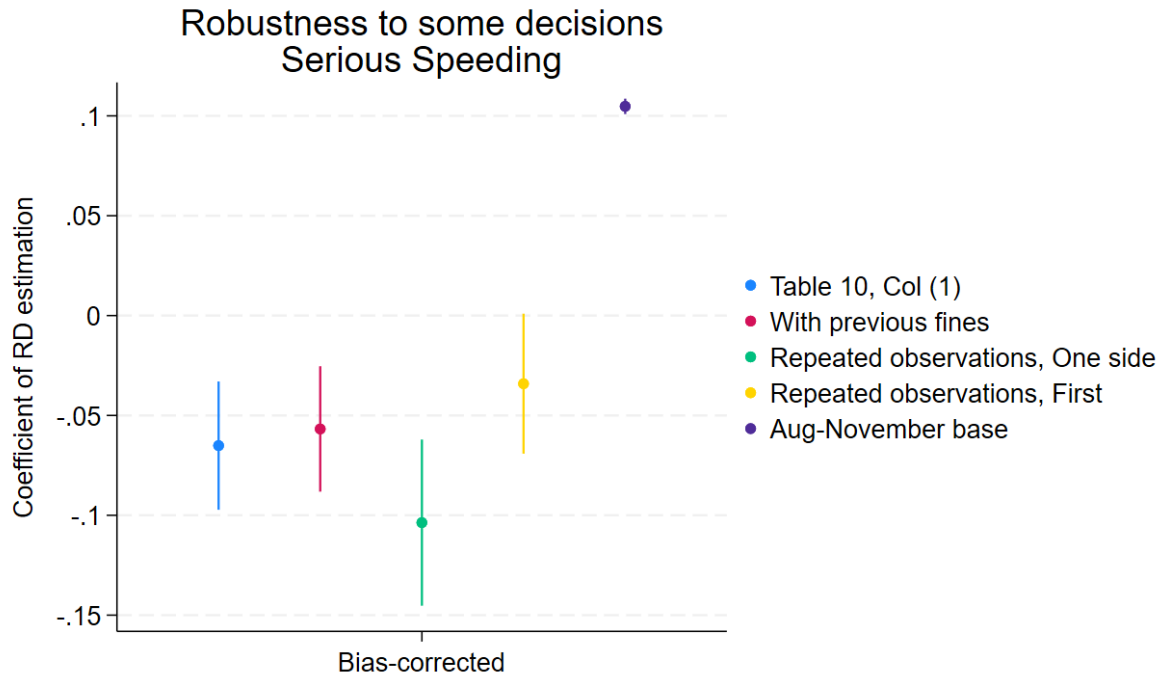


Figure 16. “Use previous fines” uses the same estimation, this time including vehicles that have had a ticket in the previous six months. “Repeated observations, One Side” shows the coefficient using Vehicles that passed more than once and always above or below the cutoff. “Repeated observations, First” shows the coefficient using Vehicles that passed more than once using the first observations as treatment assignment. “Aug-November” base makes the database bigger, by including all vehicles from August 16 to November 15, 2024.

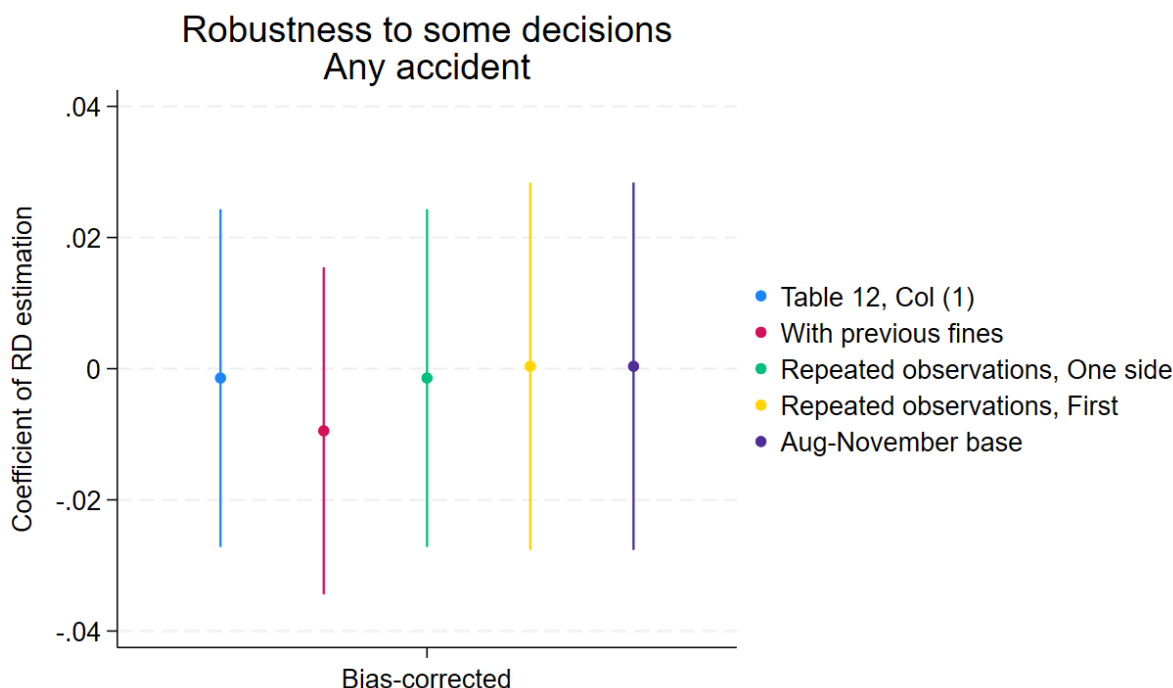


Figure 17. “Use previous fines” uses the same estimation, this time including vehicles that have had a ticket in the previous six months. “Repeated observations, One Side” shows the coefficient using Vehicles that passed more than once and always above or below the cutoff. “Repeated observations, First” shows the coefficient using Vehicles that passed more than once using the first observations as treatment assignment. “Aug-November” base makes the database bigger, by including all vehicles from August 16 to November 15, 2024.

Table 18: Effect of agent reviewing ticket. Cars

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0446** (0.0185)	-0.103*** (0.0328)	-0.0159 (0.00994)	-0.0236* (0.0138)
Mean of not ticketed	0.179	0.265	0.0451	0.0572
<i>N</i>	94536	94536	94536	94536
First stage Wald-F	65.40	65.40	65.40	65.40

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 18. Reoffending. Full sample. Cars. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one Serious speeding (above 60 km/h), and 0 otherwise. Column

4 (Times speeding>10) is a continuous variable, the total of times a vehicle was captured in Serious Speeding.

Table 19: Effect of agent reviewing ticket. Motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0235 (0.0238)	-0.0109 (0.0393)	-0.0273* (0.0159)	-0.0280 (0.0229)
Mean of not ticketed	0.123	0.176	0.0519	0.0691
<i>N</i>	38132	38132	38132	38132
First stage Wald-F	28.93	28.93	28.93	28.93

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 19. Reoffending. Full sample. Cars. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one Serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle was captured in Serious Speeding.

Table 20: Effect of agent reviewing ticket. Any accident. By quintile

	(1) Matched Sample	(2) Quintile 1	(3) Quintile 2	(4) Quintile 3	(5) Quintile 4	(6) Quintile 5
Ticket	-0.000286 (0.00667)	0.00825 (0.0329)	-0.00861 (0.0162)	-0.0128 (0.00901)	0.00541 (0.0114)	0.00699 (0.0142)
Mean of not ticketed	0.0158	0.0330	0.0132	0.0112	0.0114	0.0206
<i>N</i>	48230	3562	8772	10223	12150	13523
First stage Wald-F	46.28	3.551	8.688	13.11	12.64	13.37

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 20. Behavioral change by price quintile. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least accident, and 0 otherwise.

Table 21: Effect of receiving a ticket. Future non-Speeding tickets

	(1) Quintile 1 Other traffic tickets	(2) Quintile 2 Other traffic tickets	(3) Quintile 3 Other traffic tickets	(4) Quintile 4 Other traffic tickets	(5) Quintile 5 Other traffic tickets
Bias-corrected	-0.0205 (0.101)	-0.141** (0.0602)	-0.113* (0.0683)	-0.0520 (0.0467)	-0.00661 (0.0507)
Mean of not ticketed	0.129	0.141	0.120	0.110	0.0974
N	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Right of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21. Behavioral change. Matched Sample. Dependent variable is a dummy variable that takes a value of 1 if the vehicle ever had another ticket for any cause after the first speeding instance, and 0 otherwise.

Table 22: Effect of receiving a ticket. By month

	(1) Serious Speeding any time	(2) Serious Speeding Month 1	(3) Serious Speeding Month 2	(4) Serious Speeding Month 3	(5) Serious Speeding Month 4	(6) Serious Speeding Month 5	(7) Serious Speeding Month 6
Bias-corrected	-0.0651*** (0.0164)	-0.00864 (0.00725)	-0.0310*** (0.00825)	-0.0146* (0.00774)	0.00439 (0.00775)	-0.0359*** (0.00741)	0.00937** (0.00445)
Mean of not ticketed	0.0582	0.0102	0.0150	0.0136	0.0129	0.00994	0.00423
N	136533	136533	136533	136533	136533	136533	136533
N left of cutoff	52468	52468	52468	52468	52468	52468	52468
N right of cutoff	57605	57605	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 22. Behavioral change. Full Sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Serious speeding (above 60 km/h), and 0 otherwise. Columns 2 to 7 measure the same variable by month.

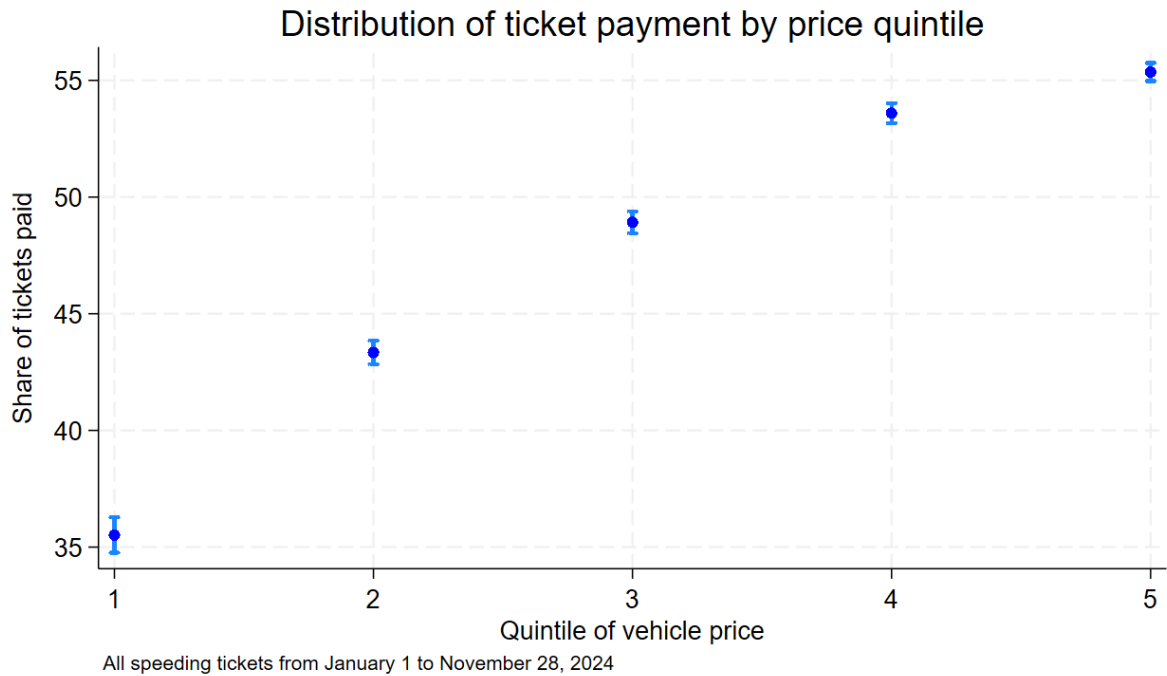


Figure 23. Ticket payment by vehicle price. Data from *Secretaría de Movilidad*.

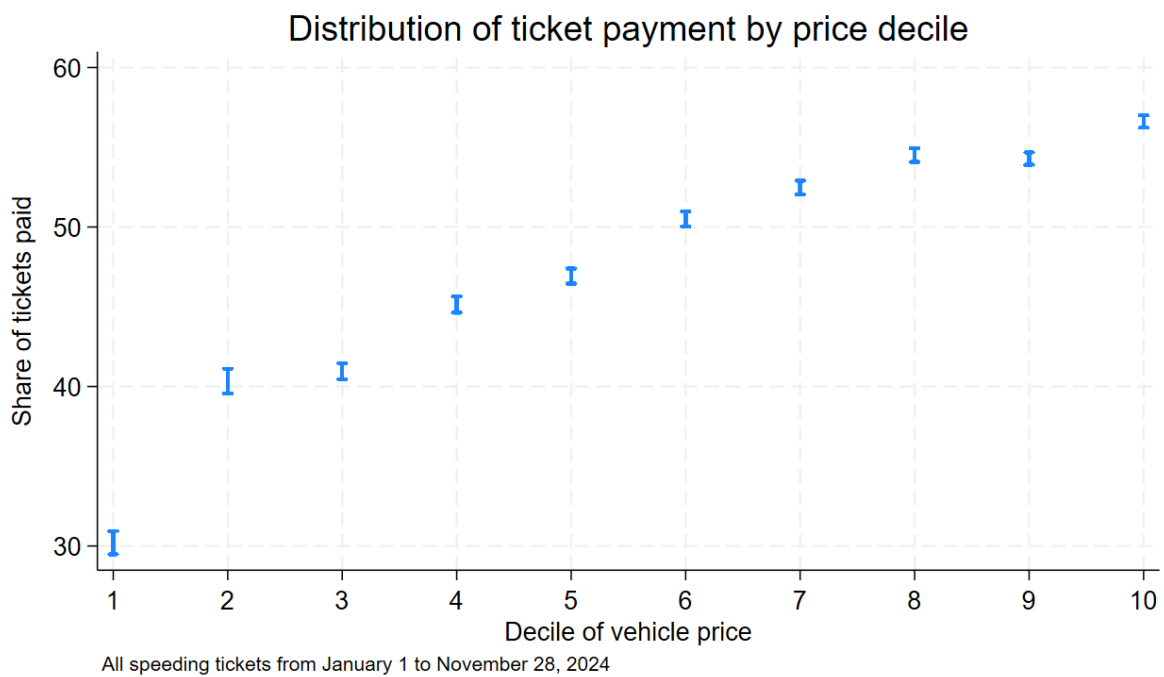


Figure 24. Ticket payment by decile of vehicle price. Data from *Secretaría de Movilidad*.

Smoothnes of observables Full sample

	(1) Hour	(2) Day	(3) Weekend	(4) Motorcycle	(5) Before scaled
Bias-corrected	0.0105 (0.0236)	-0.0185 (0.0141)	0.00330 (0.00317)	0.00245 (0.00305)	0.305 (0.309)
N	2056048	2056048	2056048	2056048	2056048
left	1573654	1573654	1573654	1573654	1573654
right	228573	228573	228573	228573	228573
bwleft	10.50	10.50	10.50	10.50	10.50
bwright	14.50	14.50	14.50	14.50	14.50
Mean	11.74	4.095	0.323	0.355	39.25

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01